

Dongfang Zhao, Ph.D.

Homepage: hpdic.github.io

Professional Preparation

Katholieke Universiteit Leuven, Belgium	Statistics, Diploma	2005
Emory University, Atlanta, GA	Computer Science, MS	2008
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
University of Washington, Seattle	Computer Science, PostDoc	2017

Employment

2023–present	Assistant Professor, University of Washington
2019–2023	Lead Developer, Moka Blox LLC, Davis, CA
2018–2020	Visiting Professor, University of California, Davis
2018–2023	Assistant Professor, University of Nevada, Reno
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer, Epic Systems Corporation, Madison, WI
2006–2007	Research Intern, Centers for Disease Control and Prevention, Atlanta, GA
2003–2004	Research Intern, Interuniversity Microelectronics Centre, Belgium

Selected Awards

2022	Federal Research Grant, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000

Selected Services

2021–present	Editorial Board	IEEE Transactions on Distributed & Parallel Systems
2023	Search Committee	One Associate Professor of Artificial Intelligence
2021	Search Committee	Two Assistant Professors of Cybersecurity
2019	Organizing Committee	ACM International Middleware Conference, UC Davis
2018–2023	Grant Review Panels	NSF CISE IIS, NSF CISE OAC, DOE ASCR

Selected Publications

(**Dongfang Zhao** is the student's *primary advisor, [†]thesis committee member, or [‡]postdoc co-author)

- [1] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data, SIGMOD 2023.
- [2] *Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data. 8(5): 1259-1272 (2022).
- [3] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [4] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [5] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [6] *Abdullah Al-Mamun, Jun Dai, Xiaohua Xu, Mohammad Sadoghi, Haoting Shen, **Dongfang Zhao**. Poster: DEAN: A Blockchain-Inspired Consensus Protocol Enabling Trustworthy Edge Computing. Network and Distributed System Security Symposium (NDSS). 2020.
- [7] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020).
- [8] [†]Heyang Qin, [†]Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [9] [‡]Parmita Mehta, [‡]Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment, VLDB 2017.
- [10] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data. 3(1): 107-117 (2017).
- [11] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016).
- [12] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Services Computing. 9(1): 96-109 (2016).
- [13] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence. 31(1): 86-98 (2009).